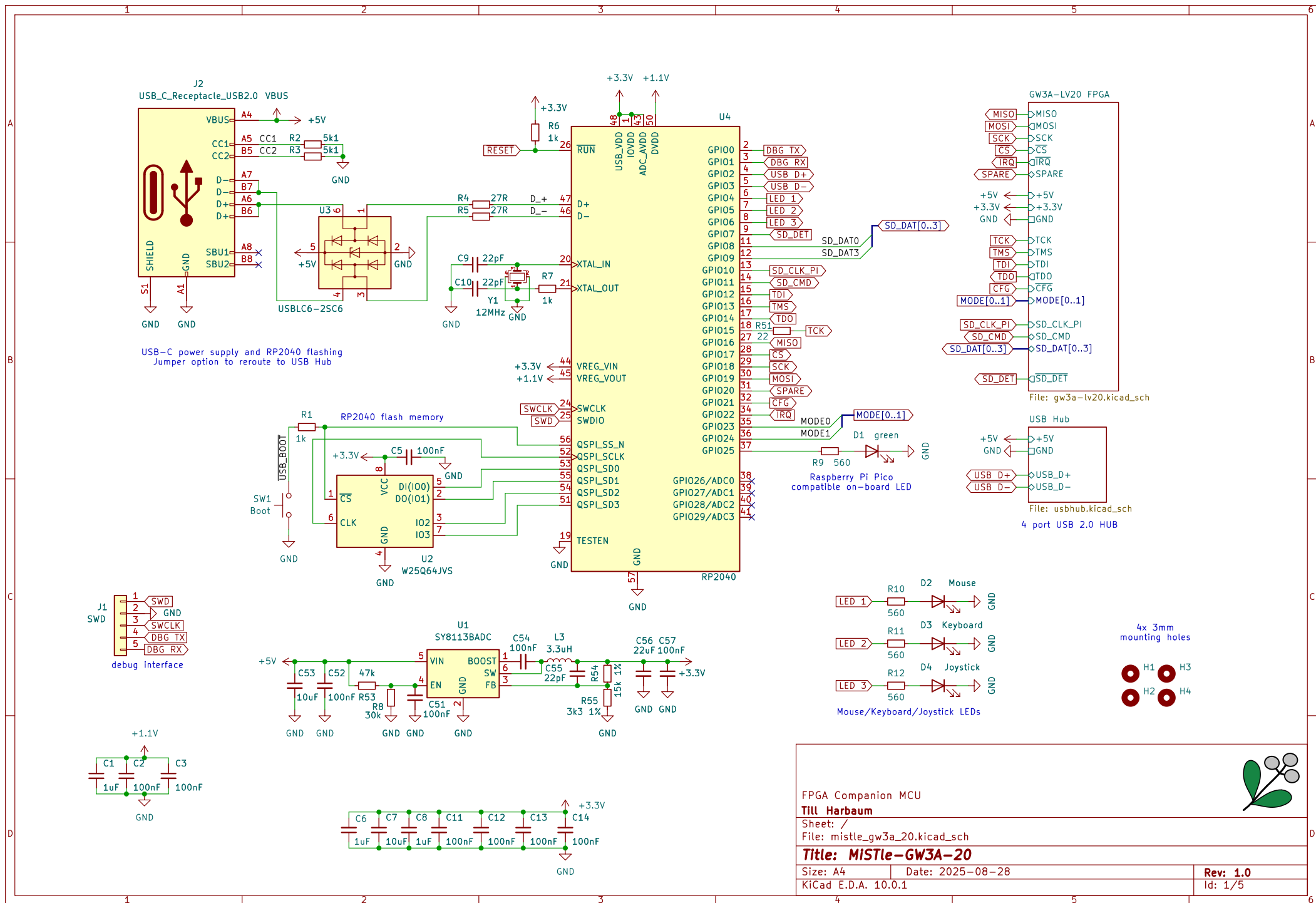




FPGA Companion MCU

Till Harbaum

Sheet: /

File: mistle_gw3a_20.kicad_sch

Title: Mistle-GW3A-20

Size: A4

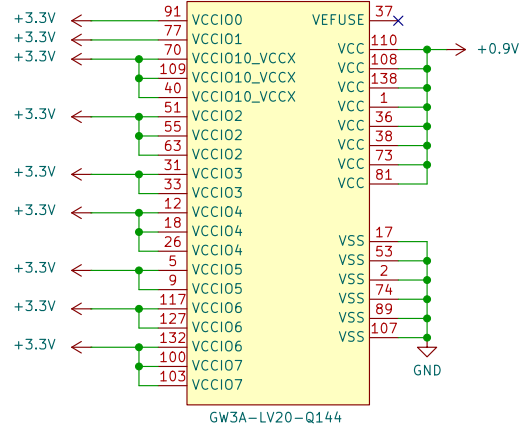
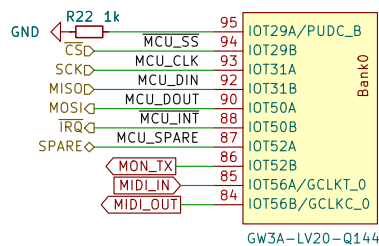
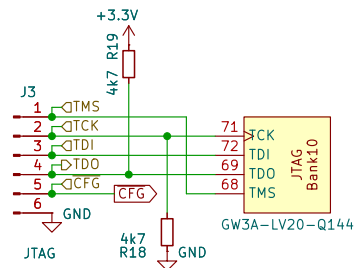
Date: 2025-08-28

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Rev: 1.0

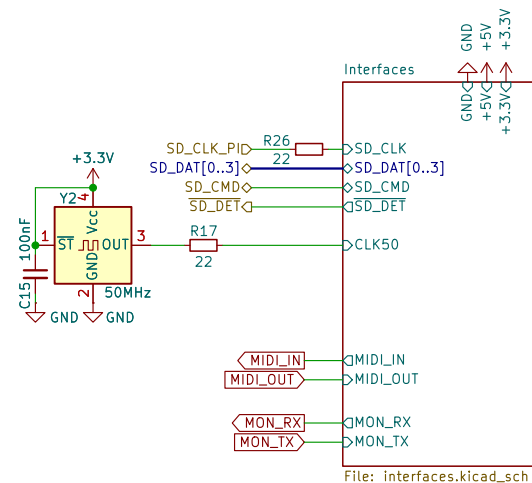
Id: 1/5

+3.3V → +3.3V
+5V → +5V
GND → GND

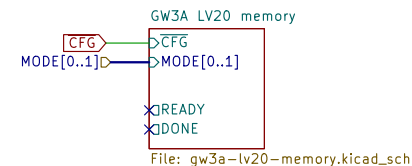


DS1351 3.1.2: eFuse voltage 1.62~1.98V if efuse writing is needed. Can be left open otherwise

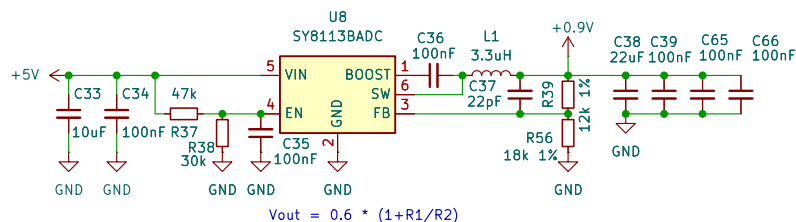
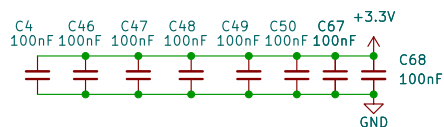
TODO: 3.1.5: What's meant by Power On Reset Voltage



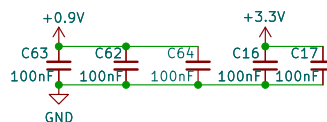
File: interfaces.kicad_sch



File: gw3a-lv20-memory.kicad_sch



$$V_{out} = 0.6 \cdot (1 + R1/R2)$$



- Pinout considerations
- MSPI WP/HOLD wired to D02 and D03 for optional QSPI
 - LVDS only available in banks 2 and 6
 - If SDRAM is powered at != 3.3V, then this voltage must also work for the flash
 - Buttons to MODE0/MODE1 as on the TN20k

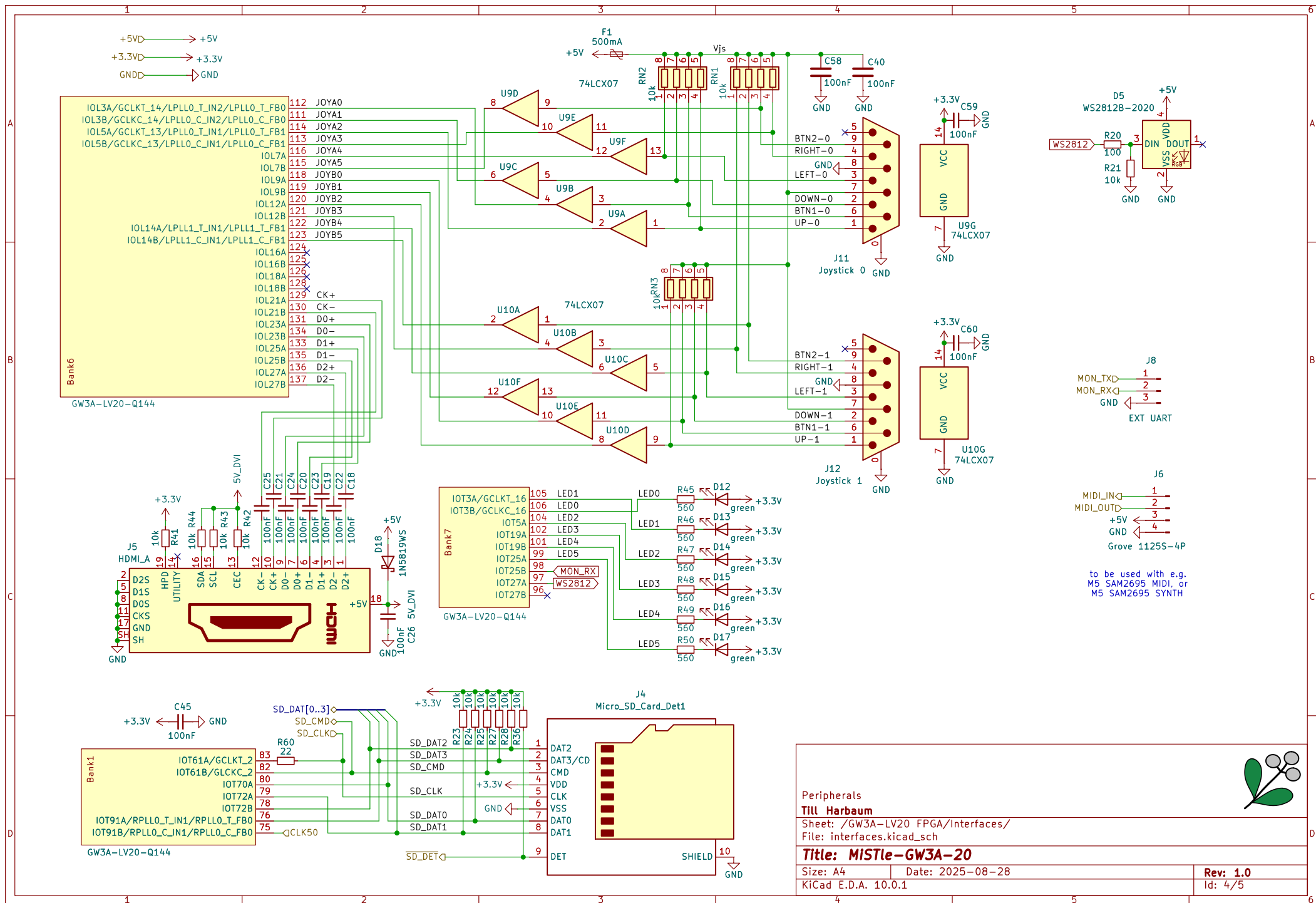
GW3A-LV20 FPGA
MISTle Project - Till Harbaum
Sheet: /GW3A-LV20 FPGA/
File: gw3a-lv20.kicad_sch

Title: MISTle-GW3A-20

Size: A4 Date: 2025-08-28
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Rev: 1.0
Id: 2/5





Peripherals

Till Harbaum

Sheet: /GW3A-LV20 FPGA/Interfaces/
File: interfaces.kicad_sch

Title: MISTle-GW3A-20

Size: A4 Date: 2025-08-28

KiCad E.D.A. 10.0.1

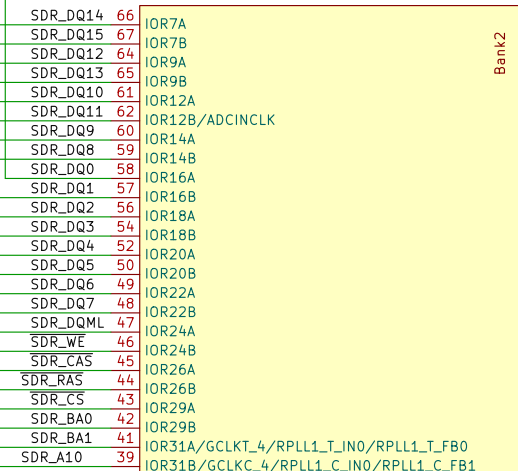
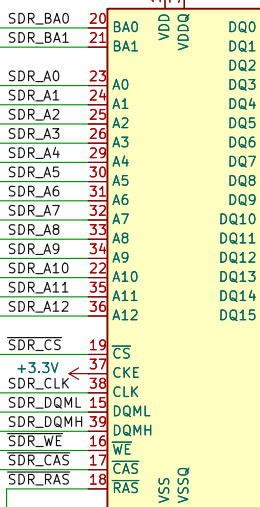
Rev: 1.0

Id: 4/5

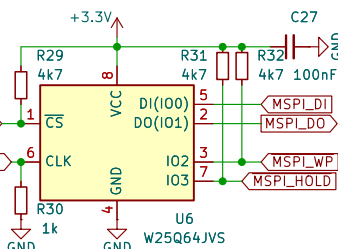


3.3V SDR SDRAM
16Mbits * 16 = 256Mbits

U13
 MT48LC16M16A2TG



GW3A-LV20-Q144



64Mbits = 8 MBytes
Flash memory

GW2AR-LV18 FPGA Memory

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Sheet: /GW3A-LV20 FPGA/GW3A LV20 memory/

File: gw3a-lv20-memory.kicad_sch

Title: MISTle-GW3A-20

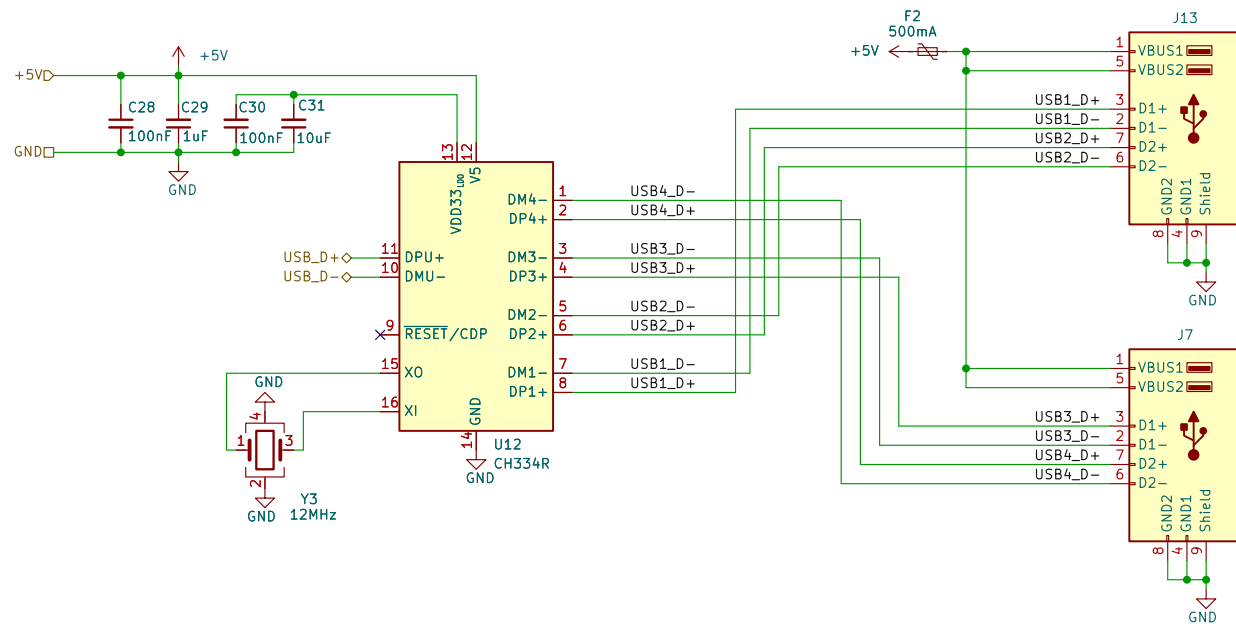
Size: A4 Date: 2025-08-28

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Rev: 1.0

Id: 5/5





USB Hub CH334R

Stefan Voss, Till Harbaum

Sheet: /USB Hub/

File: usbbhub.kicad_sch

Title: MiSTle-GW3A-20

Size: A4

Date: 2025-08-28

Rev: 1.0

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Id: 3/5

